

POLS 603 Notes*

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Contents

1	MLE setup and basics	3
1.1	Random variables and distributions	4
1.2	Likelihood function	7
1.3	Aside: Computer algebra systems	11
1.4	Properties of MLE	18
1.4.1	Hypothesis testing	32
1.5	Misspecification of the DGP	37
2	Numerical methods	41
2.1	Bisection	42
2.2	(Quasi) Newton methods	48
2.2.1	BFGS	53
2.3	Computational notes, tools, and tricks	56
2.3.1	Approximate v. real derivatives	56
2.3.2	Monte Carlo simulation	58
2.3.3	Bootstrapping	64
3	GLMs with continuous and continuous-ish data	70
3.1	Normal GLM	70
3.1.1	Normal panel models	74
3.2	Log-normal GLM	78
3.3	Poisson GLM (Count data and more)	81
3.3.1	Negative binomial GLM	84
3.3.2	Application	85
3.3.3	Vuong's Non-nested model test	101
3.4	Extensions to the Poisson GLM: instrumental variable	107
3.4.1	Two-step ML estimators	108
4	GLMs: Binary, ordered, and multinomial outcomes	108
4.1	Binomial GLM	108
4.2	Application	112
4.3	Extensions to the binomial model	112
4.3.1	Heteroskedasticity	112
4.3.2	Instrumental variables	112
4.3.3	Panel data	112

5	Duration data	112
6	Missing Data	112
7	Structural models of individual choice	112
8	Structural models of strategic interaction	112

1 MLE setup and basics

Before we start getting this semester, let's refresh some of our memory about what empirical models are, what they're good for, and what are actual goals are in learning this stuff.

In the kind of social science we do there are three main uses for empirical models:

1. **Predict** values of y for new values of X that maybe we haven't observed yet (e.g., forecasting election outcomes or predicting whether some individual enters a job training program)
2. **Describe** how do changes in y match changes in X (on average voters of color tend to vote at lower rates than white votes)
3. **Identify the treatment/marginal (causal) effect** that a change in x_i will have on values of y_i (i.e., How many more votes can a candidate get by spending another dollar, on average?)

We will often have one or more of these goals in mind when conducting research. Overwhelmingly, our goal is to identify a marginal effect or treatment effect. Often, sadly, we won't be in a position to credibly label it a casual effect, although we will talk about when it makes sense to think of it that way. In other cases, we'll be somewhere in the overlap of 2 and 3, where we acknowledge that we're describing associations, but taking care to get as close to 3 as possible.

Recall that the major obstacle in moving from 2 to 3 is **endogeneity**. Most often, we think about this as some kind of omitted variable bias, but it is broader than that. We can imagine breaking down estimation strategies into several approaches based on how they handle endogeneity concerns (i.e., how to get from 2 to 3):

1. Conditioning on the observables. This is probably the most common, but also the most suspect. Here we are just relying on having all the correct control variables in a regression or matching setting. This is why we often find ourselves somewhere between 2 & 3 and talking about association.
2. Instrumental variables. If a valid and strong instrument is available it can avoid many endogeneity problems.
3. Fixed effects. If we have panel data, we can control for unobserved heterogeneity among the units in our study. Under additional assumptions we can get to difference-in-differences designs
4. Structural approaches. Modeling the endogeneity directly (e.g., selection into the sample or strategic interactions).

This semester we will largely be exploring what are known as generalized linear models (GLMs). To preview there are pros and cons to consider when deciding if you want to use a GLM versus an ordinary linear model like you learned about last semester.

Pro GLMs take the types of values y can take on very seriously. For example, if y is Bernoulli then $\Pr(y = 1) \in [0, 1]$, but you probably saw last semester that the linear probability model (LPM) can produce nonsense estimates of this probability.

Con They often require additional assumptions on y , these assumptions are often not based on anything substantive or defensible.

Con In many cases, GLM estimates do not have closed-form solutions. This means we need to solve for them using numerical methods rather than having a go-to formula like $(X'X)^{-1}X'y$.

GLMs can be fit in any number of ways, but we will focus on the method of maximum likelihood. This method is very a powerful tool for fitting empirical models. At its core, a maximum likelihood estimator contains three parts:

1. A model of the data. This can be as simple or as complicated as needed, but without a model of where the data come from we cannot move on. Some common choices include a common named distribution or a theoretical model. You'll often see this model called the *data generating process (DGP)*. This model has parameters $\theta \in \Theta$ that we wish to estimate.
2. The observed data $y_i, i = 1, \dots, N$. These are realizations from the random variable y . We will fit the model to these data to estimate θ
3. A likelihood function $\mathcal{L} : \Theta \rightarrow \mathbb{R}$. In words, we input a guess at θ and $\mathcal{L}$ returns a number that describes how "likely" is it that the observed data y_i were generated from the assumed DGP/model if its parameters were in fact the guessed θ .

Intuitively, we want to find the values of θ that maximize the likelihood function. That is, what values of θ are the most likely for this model given the data.

You were probably been exposed to some applications of maximum likelihood estimation last semester, but let start at the very beginning with a review some of important terminology.

1.1 Random variables and distributions

A random variable is a function $y : S \rightarrow \mathbb{R}$, mapping from the **sample space** (the set of all possible outcomes from whatever it is we're doing) S into some kind of numeric outcome. For example, imagine that flip a coin N times and count the number of heads. The sample

space is then $S = \{H, T\}^N$, which is incredibly unwieldy. The random variable y maps these sequences into the numbers $1, 2, \dots, N$.

While random variables are, and always will be, functions, it is often useful to think about them in terms of outcomes of an observed experiment. The intuition here is that we have an experiment in mind and the random variable will record the outcome when we run in the future. In the present we are simply thinking about all the things that might happen when we run the experiment and how likely those different outcomes are.

When discussing random variables, we will frequently describe them in terms of the distribution over outcomes. These can be the probability of a specific outcome $\Pr(y = y_i)$ or that the outcome is within an interval $\Pr(a \leq y_i \leq b)$. Typically, we'll want to distinguish between discrete and continuous random variables.

A random variable y is discrete if its outcomes are a countable set (finite or not). The **probability mass function** (PMF) f describes the probability of each event such that

$$\begin{aligned} f(y_i) &= \Pr(y = y_i) \\ \Pr(y = A) &= \sum_{i: y_i \in A} f(y_i) \\ 1 &= \sum_i f(y_i), \forall y_i \in \text{supp}(y). \end{aligned}$$

Where $\text{supp}(y)$ is the **support** of y , which just means all values that y can take on with positive probability. Some examples include

- Bernoulli(p), $\text{supp}(y) = \{0, 1\}$
- Binomial(N, p), $\text{supp}(y) = \{0, 1, \dots, N\}$. The sum of N iid Bernoulli(p) random variables
- Poisson(λ), $\text{supp}(y) = \{0, 1, \dots\}$, which is the set of all non-negative integers $\mathbb{Z}_+$.

Consider tossing a coin once. This is a realization from a Bernoulli distribution, where with probability p , we observe a head. The PMF for the Bernoulli is a fairly simple

$$f(y_i|p) = p^{y_i}(1-p)^{1-y_i}.$$

If our experiment is to determine if the coin is fair, we may toss it N times. Each set of N tosses produces an observation y_i , which is the total number of heads. A few questions to make sure you're paying attention:

1. What kind of distribution is this? Binomial.
2. What is the probability that we get a specific sequence of y_i heads and $N - y_i$ tails? For example, all y_i heads in a row and then all tails? Well each toss is independent, so it is simply the product

$$\Pr(y_1 = 1) \dots \Pr(y_i = 1)(1 - \Pr(y_{i+1} = 1)) \dots (1 - \Pr(y_N = 1)) = p^{y_i}(1 - p)^{N - y_i}.$$

3. Ok now what is the probability of observing any sequence containing y_i heads? This is the probability of a specific sequence times the number of specific sequences that exist. So how many different ways are there to get y_i heads in N tosses? We'll save some and note that this is a solved problem, we can use the binomial or "choose" function:

$$\binom{N}{y_i} = \frac{N!}{(N - y_i)!y_i!}.$$

So the PMF of the binomial is

$$f(y_i|N, p) = \Pr(y = y_i) = \binom{N}{y_i} p^{y_i} (1 - p)^{N - y_i}.$$

When $N = 1$, this simplifies to the Bernoulli distribution.

A random variable y has a continuous distribution if there is a weakly positive function f such that for all a and b

$$\Pr(a < y \leq b) = \int_a^b f(y_i) dy_i$$

and

$$\int_{-\infty}^{\infty} f(y_i) dy_i = 1.$$

Now f is called a **probability density function** (PDF), which is similar to, but not the same as a PMF. For a continuous variable y , $\Pr(y = y_i) = 0$ for all y_i . Instead, the PDF $f(y_i)$, returns a density which can be thought of as reflecting how likely are to see something "near" y_i . Some examples include

- Uniform $U[a, b]$ where every value in the interval is equally likely. This PDF is then $f(y_i) = 1/(b - a)$, $\text{supp}(y) = [a, b]$
- Normal $N(\mu, \sigma^2)$, you've surely seen this before and will see it extensively, $\text{supp}(y) = \mathbb{R}$
- Exponential(λ), where $f(y_i|\lambda) = \lambda \exp(-\lambda y_i)$ for $\lambda > 0$ and $y > 0$. $\text{supp}(y) = \mathbb{R}_+$
- t , χ^2 , F , logistic, and many others

As in the binomial example, we can say that if we have a sequence of $y_1, \dots, y_N$ random

variables than their joint density is given as

$$f(y_1, y_2, \dots, y_N) = \prod_{i=1}^N f(y_i).$$

Finally, a **cumulative distribution function** (CDF) is as a function $F : \mathbb{R} \rightarrow [0, 1]$ such that $F(y_i) = \Pr(y \leq y_i)$. A few interesting facts about CDFs that follow from the axioms of probability are

1. F is weakly increasing $F(y_i) \leq F(y_j)$ for all $i \leq j$.
2. F does not have to be continuous, but it is continuous from the right, i.e.

$$\lim_{h \rightarrow 0^+} F(y_i + h) = F(y_i).$$

The notation $h \rightarrow 0^+$ means as h approaches 0 from the right ($h > 0$, decreasing to 0).

3. $\lim_{y_i \rightarrow \infty} F(y_i) = 1$ and
4. $\lim_{y_i \rightarrow -\infty} F(y_i) = 0$
5. $0 \leq F(y_i) \leq 1$

For discrete random variables, the CDF is defined as

$$F(y_i) = \sum_{y \leq y_i} y,$$

while for continuous random variables it is

$$F(y_i) = \int_{-\infty}^{y_i} f(y) dy.$$

Note, that this means that $f(y_i) = D_y F(y_i)$.

1.2 Likelihood function

Suppose that we believe the data come from a normal distribution with $\theta = (\mu, \sigma^2)$. Our goal is to produce the maximum likelihood estimate of θ . So what do we need? We have

1. A model, the normal distribution
2. Data, y_i

What's left? A likelihood function! Before we get to the likelihood in practice, let's think about what we want it to be in theory. We want to find the values of θ that are *most likely*

given the observed data so the problem we're looking at should be something like

$$\hat{\theta} = \operatorname{argmax}_{\theta \in \Theta} g(\theta|y),$$

where Θ is the **parameter space**, which is the set of all possible values that θ could be.

What is g though? Well maybe this is case where Bayes' rule can help us out,

$$g(\theta|y) = \frac{f(y|\theta)g(\theta)}{f(y)}.$$

Ok here we have:

1. $f(y|\theta)$ is the probability density/mass function of y from our model.
2. $g(\theta)$ is an unconditional distribution that maps from the parameter space $\Theta \rightarrow [0, 1]$. This is what's known as a **prior** distribution.
3. $f(y)$ is the unconditional probability of the observed data. Using the law of total probability we know that

$$f(y) = \int_{\Theta} f(y|\theta)g(\theta)d\theta.$$

We don't know $g(\theta)$ or $f(y)$, but we have some options. Regarding the latter, we can treat the observed data as fixed then $f(y)$ is constant. In this case it doesn't matter for the maximization problem and we can change the original problem to maximizing

$$h(\theta|y) \propto f(y|\theta)g(\theta)$$

That's progress! What do we do with the $g(\theta)$? We have two real options here

1. Take a guess at $g(\theta)$ using our best substantive knowledge (Bayesian approach)
2. Ignore $g(\theta)$ and only rely on the joint PDF/PMF of the data (maximum likelihood)

The pros and cons of working with a prior distribution are debated *ad nauseum*, and we will not take the time to hash those out, but we will point out that the second approach is in fact a special case of the first where we assume that all values $\theta \in \Theta$ are equally probable absent the model. This is a very specific kind of uninformative prior. In many cases, informative priors are not easy to justify. So, we'll admit our ignorance with respect to prior knowledge and cast aside g . This means we will only focus on the joint distribution of the sample $f(y|\theta)$.

This joint PDF is exactly how we're going to define the likelihood function. Specifically, we'll refer to the joint PDF when we want to discuss the probability of observing values of y given parameters θ , but we will switch the conditionality around to discuss how likely is a specific

parameter value given the data, this gives us

$$\mathcal{L}(\theta|y) = f(y|\theta)$$

.

Ok, we're making progress. Now we just need to be able to describe the joint distribution of the data. To do this let's formalize our first main assumption to make our life a little easier

Assumption A1 (DGP) *The data y_i , $i = 1, \dots, N$ are independent and identically distributed (iid) realizations from a distribution $f(y|\theta)$*

So with N iid realizations, the joint PDF is...?

$$\mathcal{L}(\theta|y) = \prod_{i=1}^N f(y_i|\theta).$$

Suppose that f is the normal PDF with parameters $\theta = (\mu, \sigma^2)$. The maximum likelihood estimates (MLEs) are then defined as

$$\begin{aligned}\hat{\theta} &= \operatorname{argmax}_{\theta \in \Theta} \mathcal{L}(\theta|y) \\ &= \operatorname{argmax}_{\theta \in \Theta} \prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-(y_i - \mu)^2}{2\sigma^2}\right).\end{aligned}$$

We could try to directly maximize this, but more often than not easier for us to work with

the log-likelihood function. To review, the log function has the following properties

$$\begin{aligned}
\log(xy) &= \log(x) + \log(y) \\
\log \prod_{i=1}^N x_i &= \sum_{i=1}^N \log(x_i) \\
\log(e^x) &= x \\
e^{\log(x)} &= x \\
\log(y^x) &= x \log(y) \\
\log(x/y) &= \log(x) - \log(y) \\
\log(x) &\leq x - 1 \\
\log(x) &= \text{undefined}, x < 0 \\
\log(0) &= -\infty \\
\log(x) &< 0, x \in (0, 1) \\
\log(1) &= 0 \\
\log(x) &> 0, x > 1 \\
D_x \log(x) &= 1/x > 0 \text{ Strictly increasing for } x > 0 \\
D_x^2 \log(x) &= -1/x^2 < 0 \text{ Concave function}
\end{aligned}$$

Why can we do this? 2 reasons:

1. The likelihood is the product of probabilities/densities, so long as there are no zero-probability events in the data, then the log will be well defined.
2. When looking for maxima, the first order conditions of the log-likelihood are

$$D_\theta \log(\mathcal{L}(\theta|y)) = \frac{1}{\mathcal{L}(\theta|y)} D_\theta \mathcal{L}(\theta|y),$$

by the chain rule. Note that this can only equal 0 when $D_\theta \mathcal{L}(\theta|y) = 0$, so the FOCs are preserved across this transformation. For notational ease, define the FOC as a function

$$s(y|\theta) = \sum_{i=1}^N D_\theta \log f(y_i|\theta),$$

where s is called the score function or the gradient function.

So we once again rewrite the problem:

$$\begin{aligned}\hat{\theta} &= \operatorname{argmax}_{\theta \in \Theta} \log(\mathcal{L}(\theta|y)) \\ &= \operatorname{argmax}_{\theta \in \Theta} L(\theta|y) \\ &= \operatorname{argmax}_{\theta \in \Theta} \sum_{i=1}^N -\frac{1}{2} \log(2\pi\sigma^2) - \left(\frac{(y_i - \mu)^2}{2\sigma^2} \right) \\ &= \operatorname{argmax}_{\theta \in \Theta} \sum_{i=1}^N -\frac{1}{2} \log(2\pi) - \frac{1}{2} \log(\sigma^2) - \left(\frac{(y_i - \mu)^2}{2\sigma^2} \right) \\ &= \operatorname{argmax}_{\theta \in \Theta} -\frac{N}{2} \log(2\pi) - \frac{N}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^N (y_i - \mu)^2.\end{aligned}$$

Now we just need to find the values μ and σ^2 that solve $s(y|\theta) = 0$.

1.3 Aside: Computer algebra systems

We could do this by hand, but let's take a little time to learn something else useful: computer algebra! Computer algebra comes in many different forms and packages. Mathematica is perhaps the most popular. (Un)fortunately for you, I couldn't afford it when I was younger, so I learned to use sympy instead which is a very nice python package for this sort of thing.

I recommend using Spyder for python work, it's an R Studio-like environment, available from <https://www.spyder-ide.org/download>. Python syntax is similar to R in many ways, but with a few notable differences, so let's take this slowly

There are two ways to load a package in python. First, you can write `import [PACKAGE]` as `[X]`, where `[X]` acts as prefix to denote functions from that package

```
## Load the package
import sympy as sy

##optional but makes the printed math nicer
sy.init_printing()
```

Alternatively, you can import specific functions using `from [PACKAGE] import` and then either list specific functions you want or use `*` to import everything. In this case, we could do

```
## Load the package
from sympy import *
```

```
##optional but makes the printed math nicer
init_printing()
```

In working with sympy, we start by declaring the symbols we're working with. These are any parameter and variables we use. We can declare certain traits about the symbols in this step too.

We will also declare the data y as an `IndexedBase`, meaning that we want to think of y as a vector with indexed elements. We can now translate our likelihood into symbols

```
## We first need to declare what our variables are
m = symbols("mu", real=True)
s = symbols("sigma^2", positive=True)
i, N = symbols("i, N", positive=True, integer=True)

## This will be used to sum over
y = IndexedBase('y')

## Now we can define the likelihood
L = -N*log(2*pi)/2 - N*log(s)/2 - Sum((y[i]-m)**2, (i,1,N))/(2*s)

## Take a look
L
```

$$-\frac{N \log(\sigma^2)}{2} - \frac{N \log(2\pi)}{2} - \frac{\sum_{i=1}^N (-\mu + y_i)^2}{2\sigma^2}$$

We now want to know the first derivatives. The function `diff` takes a derivative, we can also append it to an existing expression.

```
## First derivatives
FOC_mu = L.diff(m)
FOC_s = L.diff(s)

## Take a look
FOC_mu
```

FOC_s

$$-\frac{\sum_{i=1}^N (2\mu - 2y_i)}{2\sigma^2}$$
$$-\frac{N}{2\sigma^2} + \frac{\sum_{i=1}^N (-\mu + y_i)^2}{2(\sigma^2)^2}$$

Most of the time, you'd stop here to convert these to code to let R solve these conditions numerically. However, in this case we know there's an analytic solution, so we'll keep going. Unfortunately, the symbolic equation solvers don't always play well when the symbols we want to solve for inside a summation. We can work around this by doing a manual rewrite here.

```
FOC_mu = -N*m/s + Sum(y[i], (i,1,N))/s
mu_hat = solve(FOC_mu, m)
mu_hat
```

$$\left[\frac{\sum_{i=1}^N y_i}{N} \right]$$

There are also various simplification commands that could help us here. Warning: sometimes they help, sometimes they don't, sometimes you have to trial and error your way into something that looks better. Some commands you may want to know here:

- `simplify` like it says, it will do its best to make it simple.
- `expand` will expand the expression as much as it can
- `cancel` will try to cancel any common expressions in fractions
- `doit` will evaluate the expression as best it can
- `collect` will try to factor out the terms you list

There are others, but that's a good start.

```
FOC_mu = L.diff(m)
solve(FOC_mu, m)
simplify(FOC_mu)
simplify(FOC_mu).doit()
simplify(FOC_mu).expand()
simplify(FOC_mu).expand().doit()
solve(simplify(FOC_mu).expand().doit(),m)
```

$$\begin{aligned}
& \frac{\sum_{i=1}^N (-\mu + y_i)}{\sigma^2} \\
& \frac{\sum_{i=1}^N (-\mu + y_i)}{\sigma^2} \\
& \frac{\sum_{i=1}^N -\mu}{\sigma^2} + \frac{\sum_{i=1}^N y_i}{\sigma^2} \\
& -\frac{N\mu}{\sigma^2} + \frac{\sum_{i=1}^N y_i}{\sigma^2} \\
& \left[\frac{\sum_{i=1}^N y_i}{N} \right]
\end{aligned}$$

```
s2_hat = solve(FOC_s, s)
```

$$\left[\frac{\sum_{i=1}^N (\mu^2 - 2\mu y_i + y_i^2)}{N} \right]$$

Ok, so where are we here

$$\begin{aligned}
\hat{\mu}_{MLE} &= \frac{1}{N} \sum_{i=1}^N y_i \\
\hat{\sigma}_{MLE}^2 &= \frac{1}{N} \sum_{i=1}^N (y_i - \hat{\mu}_{MLE})^2.
\end{aligned}$$

Final question, how do we know it's a maximum? We would have to check the second derivatives, otherwise known as the Hessian

$$H(y|\theta) = \sum_{i=1}^N D_{\theta\theta'} \log f(y_i|\theta).$$

What do we want to find here? Negative (scalar) or negative definite (matrix). In this case, we'll look at the two scalar versions

```
D2_mu = FOC_mu.diff(m)
D2_mu
```

$$-\frac{\sum_{i=1}^N 2}{2\sigma^2}$$

Good!

Turning to the variance estimator, let's make our notation a little easier by defining a vector $e = (y_i - \hat{\mu}_{MLE})_{i=1}^N$, then $\hat{\sigma}_{MLE}^2 = e'e/N$. From here we get

```
D2_s = FOC_s.diff(s)
e = symbols("e", real=True)
D2_s = D2_s.subs([(s, e*e/N), (m, mu_hat[0])])
D2_s = D2_s.subs([(Sum((y[i]-mu_hat[0])**2, (i, 1, N)), e*e)])
D2_s
```

$$-\frac{N^3}{2e^4}$$

Also good!

Presumably you recognize the estimator for μ as the typical sample mean and for σ^2 we get a common, but not *the* common sample variance estimator. As a reminder, the typical sample variance estimator is

$$s^2 = \frac{1}{N-1} \sum_{i=1}^N (y_i - \hat{\mu})^2,$$

so what's the difference? Well for one thing the MLE is a biased estimator of σ^2 . This gives us our first result of the day

Property A1 *The method of maximum likelihood is not unbiased.*

This seems like a bit of a downer, but no worries. We will see that some ML estimators are unbiased, while others are not. We'll see other results later on to make this approach attractive.

For now, however, we can try another example. Suppose that we have $y_1, \dots, y_N$ forming a random sample from a Bernoulli distribution with parameter $\theta \in [0, 1]$.

1. What values can y_i take on? 0 or 1
2. What is the likelihood function for this problem?

$$\mathcal{L}(\theta|y) = \prod_{i=1}^N \theta_i^{y_i} (1 - \theta)^{1-y_i}$$

3. What is the log-likelihood?

$$L(\theta|y) = \sum_{i=1}^N \log(\theta) y_i + \log(1 - \theta) (1 - y_i)$$

4. What is the MLE of θ ? Start with the first order condition:

```
## We first need to declare what our variables are
t = symbols("theta", real=True, positive=True)
N = symbols("N", positive=True, integer=True, real=True)
i = symbols('i', cls=Idx)

## This will be used to sum over
y = IndexedBase('y')

## Now we can define the likelihood
L = log(t)*Sum(y[i], (i,1,N)) + log(1-t)*Sum(1-y[i], (i,1,N))

FOC = L.diff(t)
##look to make sure we're clear to solve
FOC
```

$$-\frac{\sum_{i=1}^N (1 - y_i)}{1 - \theta} + \frac{\sum_{i=1}^N y_i}{\theta}$$

```
t_hat = solve(FOC, t)
t_hat
```

$$\left[\frac{\sum_{i=1}^N y_i}{N} \right]$$

The sample mean again!

Let's try one more. Suppose we have some data $y_1, \dots, y_N$ that we believe to have come from a uniform distribution. Further suppose that we know that the lower bound of this distribution is 0, but we don't know the upper limit so our model is that $y_i \sim U[0, \theta]$. Find the MLE of θ .

1. The PDF here is

$$f(y | \theta) = \begin{cases} \frac{1}{\theta} & y \in [0, \theta] \\ 0 & \text{otherwise.} \end{cases}$$

2. The log-likelihood becomes

$$L(\theta | y) = \sum_{i=1}^N \begin{cases} -\log(\theta) & y_i \in [0, \theta] \\ -\infty & \text{otherwise} \end{cases}$$

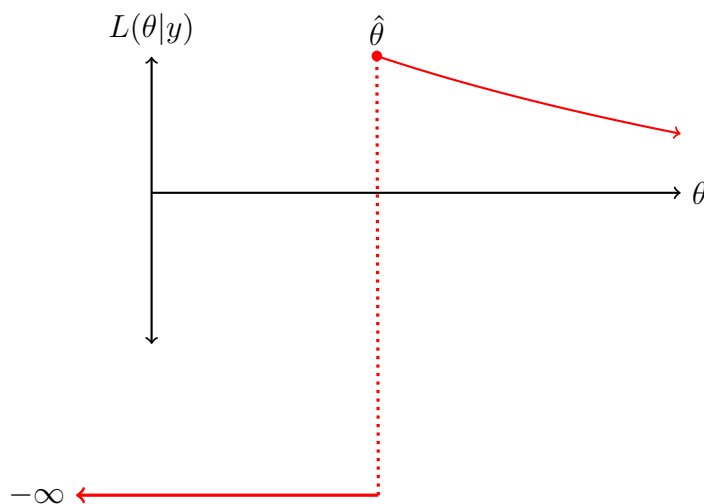
$$= \begin{cases} -N \log(\theta) & y_i \in [0, \theta], \forall i \\ -\infty & \text{otherwise} \end{cases}$$

3. The first order condition is

$$D_{\theta}L(\theta|y) = \begin{cases} -N/\theta & y_i \in [0, \theta], \forall i \\ 0 & \text{otherwise} \end{cases}$$

Note this doesn't do us a lot of good. We know that the FOC will be 0 if we pick a θ that makes the log-likelihood $-\infty$, but we also know that that will not be a maximum. For the rest of this function we get something that is strictly decreasing in θ (negative first derivative). So what value maximizes a strictly decreasing function? (The smallest value we can put into it).

Figure 1.1: Log-likelihood for $U[0, \theta]$



```
set.seed(1)
y <- runif(500, max=3)
theta <- seq(1, 5, length=10000)
L <- function(theta,y){
  return(sum(ifelse(y <= theta, -log(theta), -Inf)))
}
```

```
}
LL <- sapply(theta, L, y=y)
theta[which.max(LL)]
```

```
## [1] 2.988599
```

```
max(y)
```

```
## [1] 2.988232
```

So we want the smallest value of θ , but only to what point? We don't θ to be smaller than any of sample values!

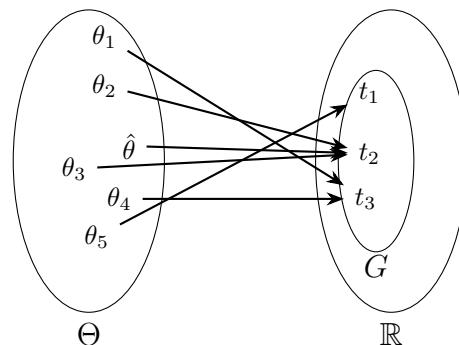
If we pick $\theta < \max(y)$ then $L(\theta | y) = -\infty$ (this is called a **zero likelihood problem**, basically it means that we have guessed a parameter value that is inconsistent with the data and model). So smallest value of θ that avoids this problem is $\max(y)$. This is the maximum likelihood estimator. So the MLE of θ is $\max(y)$. Is this biased or unbiased? What do you think happens to this bias as the sample size increases?

1.4 Properties of MLE

Now that we have some experience finding MLEs, let's talk a little bit about why we like them. We'll start with a very nice property called **invariance**.

Property A2 *If $\hat{\theta}$ is an MLE of θ , then $g(\hat{\theta})$ is an MLE of $g(\theta)$, where $g : \Theta \rightarrow \mathbb{R}$ is an arbitrary function of the parameters.*

Proof. Let G be the image of Θ under g (i.e., G is the set of values that $g(\theta)$ takes on).



For each element $t \in G$, define the restricted parameter space $G_t = \{\theta : g(\theta) = t\}$. These are

the values of θ that could produce this specific t . From the picture example we get

$$\begin{aligned}G_{t1} &= \{\theta_5\} \\G_{t2} &= \{\theta_2, \theta_3, \hat{\theta}\} \\G_{t3} &= \{\theta_4, \theta_1\}\end{aligned}$$

Now we can write the restricted maximization problem in terms of t

$$\begin{aligned}L_R(t) &= \max L(\theta|y) \\s.t. \quad &g(\theta) = t.\end{aligned}$$

Here $L_R(t)$ is a function that takes in t and returns the log-likelihood value that solves this constrained optimization problem. Let $\hat{t}$ be an MLE of $g(\theta)$, this means that

$$L_R(\hat{t}) = \max_{t \in G} L_R(t).$$

We now need to show that $\hat{t} = g(\hat{\theta})$ satisfies this condition. Two facts that are important for this:

1. $L_R(t)$ is the maximum of $L(\theta|y)$ over a subset of Θ
2. $\hat{\theta}$ maximizes $L(\theta|y)$ over all of Θ

Together, these facts mean that $L_R(t) \leq L(\hat{\theta}|y)$ for all $t \in G$. In other words, if you give me a t , I can give you $L_R(t)$, and we know that this value is at most $L(\hat{\theta}|y)$.

Consider $\hat{t} = g(\hat{\theta})$ (t_2 in the picture). This places $\hat{\theta} \in G_{\hat{t}}$. Since $\hat{\theta}$ maximizes L over all of Θ , it must also be the maximum over this subset, and so $L_R(\hat{t}) = L(\hat{\theta}|y)$, which as we previously showed is the maximum value L_R can take on. $\square$

In practice this means that once we estimate the parameters with maximum likelihood we can use those estimates to form other estimates, which will also be maximum likelihood estimates. Some common examples,

1. The MLE of the standard deviation is the square root of the MLE of the variance
2. If we need to estimate a quantity that we know to be strictly positive (constrained optimization), we can estimate log of it (unconstrained optimization) and then un-log it to get the MLE.
3. When we get to fitting nonlinear regression models with MLE, literally every marginal effect or predicted value

Let's return to the Bernoulli example above where we have $y_1, \dots, y_N$ forming a random sample from a Bernoulli distribution with parameter $\theta \in [0, 1]$. We said that the MLE of θ is the sample mean. What is the MLE of the variance of this distribution?

- Recall that for a Bernoulli, the variance is $\theta(1 - \theta)$, and so the MLE of the variance is $\hat{\theta}(1 - \hat{\theta})$.

The second property of MLE that we want to show is consistency. We already know that MLEs are not guaranteed to be unbiased (some are, some aren't), but maybe we want to know that they're asymptotically unbiased at least.

As a reminder, we say that an estimator $\hat{\theta}$ is consistent for parameter θ if $\hat{\theta} \xrightarrow{P} \theta^*$. This notation is read as saying that $\hat{\theta}$ **converges in probability** to θ^* , which means that for some $\varepsilon > 0$ and some $\delta > 0$ there is a sample size N such that for all $N' > N$ $\Pr(|\hat{\theta} - \theta^*| > \varepsilon) < \delta$. In words, this says that as N increases, the difference between $\hat{\theta}$ and θ^* is vanishingly small.

We'll need a few additional assumptions here:

Assumption A2 *The parameter space Θ is compact (closed and bounded).*

This is often ignored in practice, but it makes our proofs a little easier. To help you sleep at night, just imagine we set some imaginary large and silly bounds on the parameter space ahead of time that are far away from any reasonable guess at the truth.

To remind ourselves:

Bounded Every point in the set is within a certain distance of all the others. It has upper and lower bounds in each dimension

Closed A closed set is one contains its boundary points, and every sequence/limit of points within the set that converges does so to a point in the set (no holes).

Some examples:

1. The interval $[0, 1]$ is closed and bounded
2. The interval $[0, \infty)$ is closed but not bounded
3. The interval $(0, \infty)$ is not bounded and not closed
4. The interval $(0, 1)$ is bounded, but not closed

Assumption A3 *The density $f(y; \theta)$ is continuously differentiable in θ for all $y \in \text{supp}(y)$ and the support of y does not change with θ . The expected value $E[\log f(y|\theta)]$ exists.*

Property A3 *Under Assumptions A1-A3, an MLE $\hat{\theta}$ exists.*

Assumption A4 (Identification) *If $\theta_1 \neq \theta_2$, then $f(y; \theta_1) \neq f(y; \theta_2)$*

Basically, this means that for any two different parameter values, the distributions will be different. Think about the normal distribution, any different value of μ produces a different normal distribution. Basically, this says for any two guesses of θ we can say what makes them different. This assumption can fail in cases where changing θ_1 can be offset by changing θ_2 . For example, in ideal point estimation, a common problem is multiplying everyone by -1 will produce identical results, violating this assumption.

Assumption A5 *Additional technical conditions to avoid zero likelihood problems and apply the uniform law of large numbers.*

These are sometimes referred to as regularity conditions for maximum likelihood estimation. Nearly everything we do going forward will satisfy these. However, just because a model doesn't satisfy these conditions, doesn't mean that it's not consistent or anything else, just that we can't use these upcoming results out of the box.

Theorem 1 (Uniform LLN for MLE) *Under Assumptions A1-A5,*

$$\sup_{\theta \in \Theta} \left| \frac{1}{N} \sum_{i=1}^N \log f(y_i | \theta) - E[\log f(y | \theta)] \right| \xrightarrow{p} 0.$$

In case you forgot, the supremum is the smallest possible upper bound of a set. If the supremum is in the set, it is also the maximum. The ULLN is important for us, because it allows us to talk about convergence of a random function rather than just a sequence of random variables.

Property A4 *Under Assumptions A1-A5, maximum likelihood estimates are consistent.*

Before proving this we will remind ourselves of two useful facts. The first is the law of the unconscious statistician.

Theorem 2 (The Law of the Unconscious Statistician) *Let X be a random variable with pmf/pdf f and $E[X]$ is finite. For any real valued function g*

$$E[g(X)] = \sum_x g(x)f(x)$$

for discrete X and

$$E[g(X)] = \int_x g(x)f(x)dx$$

for continuous X .

The second is that for positive x , $\log(x) \leq x - 1$ and this inequality is strict everywhere except $x = 1$.

Proof. To prove Property A4, we will proceed in three steps. First, we will show that θ^* is a maximum of the population log-likelihood. Second, we will show that θ^* is unique in maximizing the population log-likelihood. Finally, we will show that the sample log-likelihood converges to the population log-likelihood and so the MLE converges to

Step 1 We start by defining the sample and population likelihoods:

$$\begin{aligned} L_N(\theta; y_i) &= \frac{1}{N} \sum_{i=1}^N \log f(y_i|\theta) \\ L^*(\theta) &= E[\log f(y|\theta)] \\ &= \int_{-\infty}^{\infty} f(y|\theta^*) \log f(y|\theta) dy \quad \text{Law of the unconscious statistician.} \end{aligned}$$

The first thing we want to show is that θ^* is a maximizer of L^* . The FOC here is

$$\begin{aligned} D_\theta L^*(\theta) &= D_\theta \int_{-\infty}^{\infty} f(y|\theta^*) \log f(y|\theta) dy \\ &= \int_{-\infty}^{\infty} D_\theta f(y|\theta^*) \log f(y|\theta) dy \quad \text{Leibniz rule} \\ &= \int_{-\infty}^{\infty} f(y|\theta^*) \frac{D_\theta f(y|\theta)}{f(y|\theta)} dy. \end{aligned}$$

We would like this to be 0 at $\theta = \theta^*$,

$$\int_{-\infty}^{\infty} f(y|\theta^*) \frac{D_\theta f(y|\theta^*)}{f(y|\theta^*)} dy = D_\theta \int_{-\infty}^{\infty} f(y|\theta^*) dy = D_\theta 1 = 0$$

So θ^* is indeed an optimum. We'll skip, but assert, the second order condition for now (maybe it'll be in a problem set or exam).

Step 2 We now want to show that for $\theta \neq \theta^*$,

$$L^*(\theta) < L^*(\theta^*).$$

$$\begin{aligned}
L^*(\theta) - L^*(\theta^*) &= \mathbb{E} [\log f(y|\theta) - \log f(y|\theta^*)] \\
&= \mathbb{E} \left[\log \frac{f(y|\theta)}{f(y|\theta^*)} \right] \\
&\leq \mathbb{E} \left[\frac{f(y|\theta)}{f(y|\theta^*)} - 1 \right] \\
\mathbb{E} \left[\frac{f(y|\theta)}{f(y|\theta^*)} - 1 \right] &= \int \left(\frac{f(y|\theta)}{f(y|\theta^*)} - 1 \right) f(y|\theta^*) dy \\
&= \int f(y|\theta) - f(y|\theta^*) dy \\
&= \int f(y|\theta) dy - \int f(y|\theta^*) dy \\
&= 1 - 1 = 0.
\end{aligned}$$

This establishes that $L^*(\theta) - L^*(\theta^*) \leq 0$, to make that strict, note that by Assumption A1 (identification), $f(\theta) \neq f(\theta^*)$ unless $\theta = \theta^*$

Step 3 We are now ready to show the last part which is that the sample likelihood function converges to the population likelihood. The ULLN does all the work for us here under our assumptions,

$$\sup_{\theta \in \Theta} |L_N(\theta|y) - L^*(\theta)| \xrightarrow{P} 0.$$

This means that as $N \rightarrow \infty$, the sample likelihood gets closer to the population likelihood and so the MLE obtained by maximizing it $\hat{\theta}_N$ will be closer to the MLE obtained by maximizing the population function, which as we showed about is θ^* .

Another way to see that is:

$$\begin{aligned}
L^*(\theta^*) - L_N(\hat{\theta}_N) &= \mathbb{E} [\log f(y|\theta^*)] - \frac{1}{N} \sum_{i=1}^N \log f(y_i|\hat{\theta}_N) \\
\frac{1}{N} \sum_{i=1}^N \log f(y_i|\hat{\theta}_N) &\xrightarrow{P} \mathbb{E} [\log f(y|\hat{\theta})] && \text{ULLN} \\
\mathbb{E} [\log f(y|\hat{\theta})] &= \mathbb{E} [\log f(y|\theta^*)] && \text{Previous steps} \\
L^*(\theta^*) - L_N(\hat{\theta}_N) &\xrightarrow{P} \mathbb{E} [\log f(y|\theta^*)] - \mathbb{E} [\log f(y|\theta^*)] && \text{Slutsky} \\
&= 0.
\end{aligned}$$

□

This is good news. So far we've seen that the finite sample properties are questionable (maybe it's biased, maybe it's not), but we now at least know that this bias goes to 0 as N increases.

Assumption A6 *The true value of θ is interior to Θ .*

Assumption A7 *The density $f(y; \theta)$ is thrice differentiable with respect to θ .*

Assumption A8 *The Fisher information matrix*

$$I(\theta) = E[(D_\theta \log f(y|\theta))(D_\theta \log f(y|\theta))'],$$

exists and is finite for all y such that $f(y|\theta) > 0$

These additional assumptions will allow us to characterize the large sample variance and distribution of MLEs. As before, nearly every example you come across will satisfy these conditions, but there is at least one example we looked at before that didn't. What was it? Estimating θ in the $U[0, \theta]$ case. Here, θ was not an interior solution, it was the edge of Θ , in this case we would need to derive the variance and distribution from the estimator itself rather than these results.

We have introduced this new matrix $I(\theta)$, which we're calling the Fisher information; what is it? In short, $I(\theta)$ tells us, on average, how informative each piece of data is at estimating θ . The more informative a sample is, the smaller the variance should be. That is, as $I(\theta)$ increases, the variance of the estimator will decrease. Notice that the definition is just the variance of the first order conditions, so the more variance we have in the first derivatives, the less certain we are that we're at the "true" maximum, the less informative the data are. Additionally, note that in an iid sample of size N , the total information we have is thus $NI(\theta)$. We can rewrite the information as a function of second derivatives (Hessian), to make that more clear.

Result 1 (Information matrix equality) *Under assumptions A3-A8, the Fisher information $I(\theta)$ can also be written as*

$$I(\theta) = -E[D_{\theta\theta'} \log f(y|\theta)].$$

Intuitively, this means that the Fisher information is reflected in the curvature of the log-likelihood. The flatter it is, the less information we have. The more pointy it is the more certain we are that we got a good answer.

We can be a little more precise about this relationship between variance and information. Consider a random sample $y_1, \dots, y_N$ from a distribution $f(y|\theta)$ and an arbitrary estimator $\hat{\theta}_A$ with $\text{Var}(\hat{\theta}_A) < \infty$.

To be clear, we're looking to say something about $\text{Var}(\hat{\theta}_A)$, and we'll rely on the **Cauchy-Schwarz inequality** to say it. The inequality is

$$\text{Var}(\hat{\theta}_A) \geq \text{Cov}(s(\theta|y), \hat{\theta}_A) \text{Var}(s(\theta|y))^{-1} \text{Cov}(\hat{\theta}_A, s(\theta|y)).$$

So these we need to describe the two right-hand terms. The variance is the easier one. As we saw above, it's the variance of the gradients which is just the total Fisher information in the sample:

$$\text{Var}(s(\theta|y)) = NI(\theta).$$

The other term

$$\begin{aligned} \text{Cov}(s(\theta|y), \hat{\theta}_A) &= E[s(\theta|y)\hat{\theta}'_A] - E[s(\theta|y)] E[\hat{\theta}_A]' \\ &= E[s(\theta|y)\hat{\theta}'_A] \\ &= \int_y \dots \int_y f(y|\theta) s(\theta|y) \hat{\theta}'_A dy_1 \dots dy_N \quad \text{Law of US} \\ &= \int_y \dots \int_y D_\theta f(y|\theta) \hat{\theta}'_A dy_1 \dots dy_N. \end{aligned}$$

Note that also by Law of the Unconscious Statistician we get

$$E[\hat{\theta}_A] = \int_y \dots \int_y f(y|\theta) \hat{\theta}_A dy_1 \dots dy_N.$$

Taking the derivative of this with respect to θ we get

$$D_\theta E[\hat{\theta}_A] = \int_y \dots \int_y D_\theta f(y|\theta) \hat{\theta}_A dy_1 \dots dy_N = \text{Cov}(s(\theta|y), \hat{\theta}_A).$$

So now we can go back to the inequality to get

Result 2 (Cramér-Rao lower bound or the information inequality) *Under assumptions, A6-A8, the lower bound on the variance of an estimator is given as:*

$$\text{Var}(\hat{\theta}_A) \geq \left(D_\theta E[\hat{\theta}_A] \right) [NI(\theta)]^{-1} \left(D_\theta E[\hat{\theta}_A] \right).$$

This result tells us that the lowest variance we can get for any estimator of θ , under our regularity conditions, is a function of the estimator's expected value, the sample size, and the Fisher information. Now suppose that the estimator is unbiased, then the above simplifies, with

$$D_\theta E[\hat{\theta}_A] = D_\theta \theta = I,$$

and then

$$\text{Var}(\hat{\theta}_A) \geq [NI(\theta)]^{-1}.$$

So under our regularity conditions, the smallest variance we can get for an unbiased estimator is the inverse of the Fisher information for the whole sample. Estimators that obtain the CR lower bound given a fixed expected value and sample size are the most efficient estimators (i.e., they use all the available Fisher information).

Before we move into the next result, we need to remind ourselves about Taylor series expansions and the mean value theorem. For ease, I'll present them in scalar notation, but they extend to the matrix cases we'll use.

Theorem 3 (Mean value theorem) *Let $f : [x_1, x_2] \rightarrow \mathbb{R}$ be a continuous, differentiable function, then there exists a point $\bar{x} \in (x_1, x_2)$ such that*

$$D_x f(\bar{x}) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

and a version of Taylor's theorem

Theorem 4 (Taylor's theorem) *Let $k \in \mathbb{Z}$ and let $f : \mathbb{R} \rightarrow \mathbb{R}$ be $k + 1$ -differentiable at some point $a \in \mathbb{R}$. Then*

$$f(x) = \sum_{j=0}^k \frac{D_x^j f(a)}{j!} (x - a)^j + R_{k+1}(x).$$

We group these together because they are intimately related. To see this, let $x_1 = a$ and $x_2 = x$ and rewrite the mean value theorem to be

$$\begin{aligned} D_x f(\bar{x}) &= \frac{f(x) - f(a)}{x - a} \\ f(x) &= f(a) + D_x f(\bar{x})(x - a) \end{aligned}$$

which is a Taylor expansion with $k = 0$ and $R_0(x) = Df(\bar{x})(x - a)$. It turns out we can generalize this combo by letting

$$R_{k+1}(x; \bar{x}) = \frac{D^{k+1} f(\bar{x})}{(k+1)!} (x - a)^{k+1}.$$

The main thing to note here is that we don't know, and often don't need to know what $\bar{x}$ is. Just that it exists, which is to say there will be such a value between x and a .

For our purposes, we will often be working with the log-likelihood function and setting x to be true but unknown value, a to be the MLE, and $\bar{x}$ to just be something between these two. In this case, as a converges to x by the consistency of MLE, $\bar{x}$ also converges to x faster, making the remainder convergence to 0. Let's illustrate these results with the binomial likelihood:

```
set.seed(123)
theta <- .3 # Truth
N <- 1000 # Sample size

## Data
y <- rbinom(N, size = 1, prob=theta)

## functions
L <- function(theta, y){ #log likelihood
  return(sum(dbinom(y, size=1,prob=theta, log=TRUE)))
}
s <- function(theta, y){ #score
  return(sum((y-theta)/(theta*(1-theta))))
}
H <- function(theta, y){ #hessian
  top <- -theta^2+2*theta*y-y
  bottom <- theta^2*(theta^2-2*theta+1)
  return(sum(top/bottom))
}
DH <- function(theta, y){ #third deriv
  top <- 2*(theta^3 - 3*theta^2*y + 3*theta*y - y)
  bottom <- theta^3*(theta^3 - 3*theta^2 + 3*theta - 1)
  return(sum(top/bottom))
}
T1 <- function(theta, y,a){ #first order Taylor
  return(L(a, y)+ (theta-a)*s(a,y))
}
T2 <- function(theta, y,a){ #second order Taylor
  return(T1(theta, y,a) + ((theta-a)^2 * H(a,y))/2)
}

theta.hat <- mean(y) #MLE of theta
```

```
theta.hat
```

```
## [1] 0.295
```

```
Theta <- seq(0.1,.4,length=5000) #Range of theta's
```

```
Lout <- sapply(Theta, L, y=y)
```

```
T1.out <- sapply(Theta, T1, y=y, a=theta.hat)
```

```
T2.out <- sapply(Theta, T2, y=y, a=theta.hat)
```

```
par(mfrow=c(1,2))
```

```
plot(Lout~Theta, type='l',  
      xlab=expression(theta),  
      ylab="LL")
```

```
lines(T1.out~Theta, col='red')
```

```
lines(T2.out~Theta, col='blue')
```

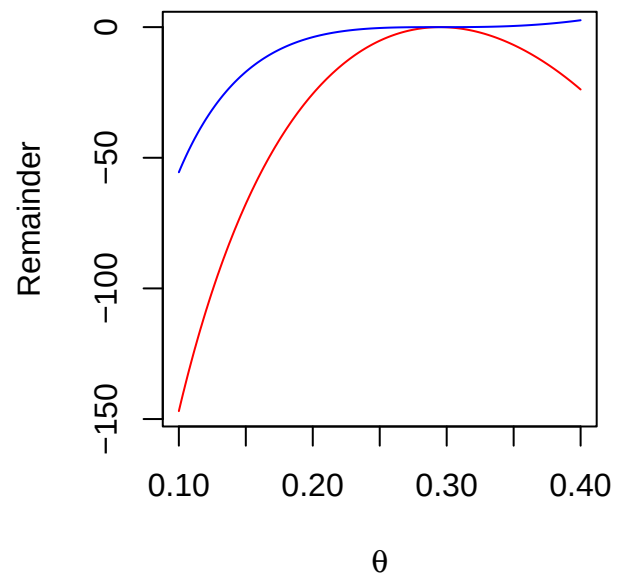
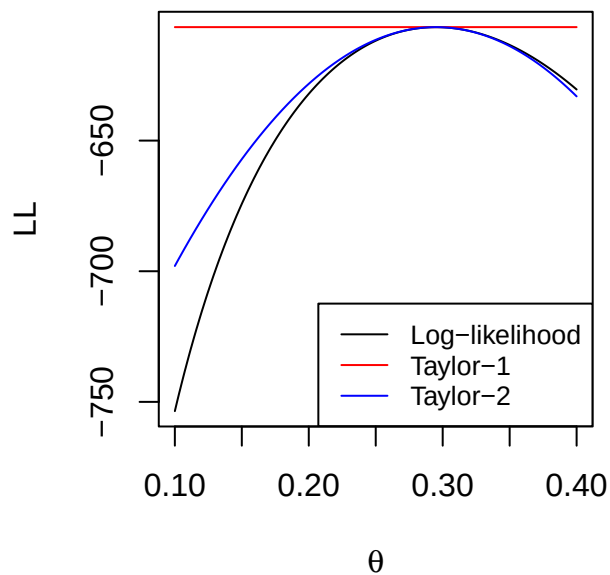
```
legend("bottomright",  
       lty=1,  
       c("Log-likelihood",  
         "Taylor-1",  
         "Taylor-2"),  
       cex=.8,  
       col=c("black", "red", "blue"))
```

```
R1 <- Lout -T1.out
```

```
R2 <- Lout -T2.out
```

```
plot(R1~Theta, type='l',  
      ylab="Remainder",  
      col='red', xlab=expression(theta))
```

```
lines(R2~Theta, col='blue')
```



```
## Why is the first order approx flat? Because  $s(\theta; y)=0$ 
```

```
## Strong improvement from first order to second order
```

```
## Let's look at true theta
```

```
## polynomial approx
```

```
c(L(theta,y),
  T1(theta, y, theta.hat),
  T2(theta, y, theta.hat))
```

```
## [1] -606.6278 -606.5681 -606.6282
```

```
## remainder at this point
```

```
c(L(theta,y)- T1(theta, y, theta.hat),
  L(theta,y)-T2(theta, y, theta.hat))
```

```
## [1] -0.0597148737  0.0003885041
```

```
## Illustrating the mean value form of the remainder
```

```
## Set the range of  $\bar{\theta}$ :  $[\theta, \hat{\theta}(\theta)]$ 
```

```
Theta.bar <- seq(theta, theta.hat, length=1000)
```

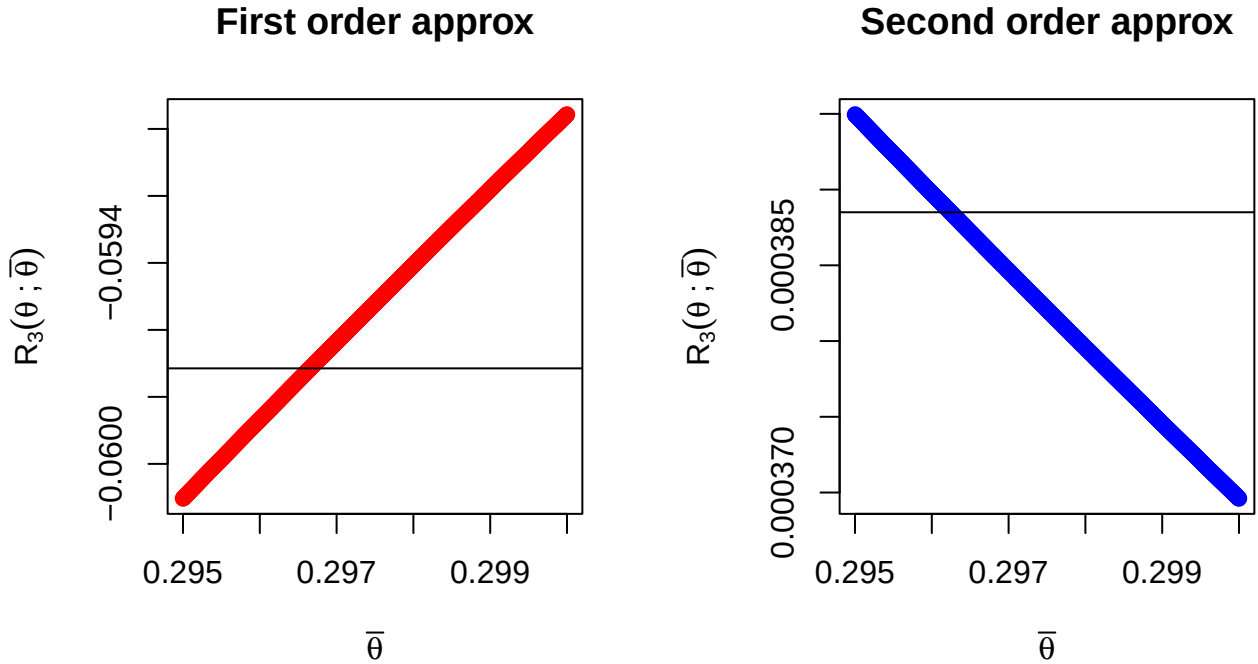
```
par(mfrow=c(1,2),mar=c(5, 4.5, 4, 2) + 0.1)
```

```

plot(sapply(Theta.bar,
           \(x){((theta-theta.hat)^2 * H(x,y))/2})~Theta.bar,
     ylab=expression(R[3](theta~";"~bar(theta))),
     xlab=expression(bar(theta)),
     main="First order approx",
     col="red")
abline(h=L(theta,y)- T1(theta, y, theta.hat))

plot(sapply(Theta.bar,
           \(x){((theta-theta.hat)^3 * DH(x, y=y))/6})~Theta.bar,
     ylab=expression(R[3](theta~";"~bar(theta))),
     xlab=expression(bar(theta)),
     main="Second order approx",
     col="blue")
abline(h=L(theta,y)- T2(theta, y, theta.hat))

```



We are now ready to present the next property of MLEs,

Property A5 *Under Assumptions A1-A8, the MLE $\hat{\theta}_N$ has an asymptotic distribution*

$$\sqrt{N}(\hat{\theta}_N - \theta^*) \xrightarrow{d} N(0, I(\theta^*)^{-1})$$

Proof. We begin with a mean-value characterization of $s(\hat{\theta}_N|y)$ around the true value θ^* , with $\bar{\theta}$ between $\hat{\theta}_N$ and θ^* . Note that because $\bar{\theta}$ is between these two terms, it will converge to θ^* as N increases

$$\begin{aligned}
s(\hat{\theta}_N|y) &= s(\theta^*|y) + H(\bar{\theta}|y)(\hat{\theta}_N - \theta^*) \\
0 &= s(\theta^*|y) + H(\bar{\theta}|y)(\hat{\theta}_N - \theta^*) && \text{FOC} \\
(\hat{\theta}_N - \theta^*) &= \left[-H(\bar{\theta}|y)\right]^{-1} s(\theta^*|y) && \text{Rearrange} \\
\sqrt{N}(\hat{\theta}_N - \theta^*) &= \left[-\frac{1}{N}H(\bar{\theta}|y)\right]^{-1} \sqrt{N}\frac{1}{N}s(\theta^*|y) \\
\frac{1}{N}H(\bar{\theta}|y) &\xrightarrow{p} I(\theta^*) && \text{LLN} \\
\frac{1}{N}s(\theta^*|y) &\xrightarrow{p} 0 && \text{LLN} \\
\sqrt{N}\left(\frac{1}{N}s(\theta^*|y)\right) &\xrightarrow{d} N(0, I(\theta^*)) && \text{CLT} \\
\sqrt{N}(\hat{\theta}_N - \theta^*) &\xrightarrow{d} N\left(0, I(\theta^*)^{-1}I(\theta^*)I(\theta^*)^{-1}\right) && \text{Slutsky/CMT} \\
\sqrt{N}(\hat{\theta}_N - \theta^*) &\xrightarrow{d} N\left(0, I(\theta^*)^{-1}\right).
\end{aligned}$$

Which is what we wanted to show. □

Immediately from this we get our next property:

Property A6 *Under Assumptions A1-A8, the MLE $\hat{\theta}$ is asymptotically efficient compared to any consistent estimator of θ .*

This follows from the fact that the MLE

1. Is consistent (asymptotically, unbiased)
2. Has an asymptotic variance $\frac{1}{N}I(\theta)^{-1}$, which is the CR lower-bound for an unbiased estimator.

So now we have that among all consistent estimators, none can be more efficient than the MLE. This is one of the main justifications for using MLE.

To estimate the variance under these assumptions, we'll typically just evaluate the information

matrix at the MLE in one of three ways

$$\begin{aligned}
\widehat{\text{avar}}(\hat{\theta}) &= \frac{1}{N} I(\hat{\theta})^{-1} \\
\widehat{\text{avar}}(\hat{\theta})_E &= -\frac{1}{N} E[D_{\theta\theta'} \log f(y|\hat{\theta})]^{-1} && \text{Exp of hessian at MLE} \\
\widehat{\text{avar}}(\hat{\theta})_H &= -H(\hat{\theta}|y)^{-1} = -\left[\sum_{i=1}^N D_{\theta\theta'} \log f(y_i|\hat{\theta}) \right]^{-1} && \text{Inv. of minus Hessian} \\
\widehat{\text{avar}}(\hat{\theta})_{OPG} &= \left[\sum_{i=1}^N \left[D_{\theta} \log f(y_i|\hat{\theta}) \right] \left[D_{\theta} \log f(y_i|\hat{\theta}) \right]' \right]^{-1} && \text{Outer product of gradients}
\end{aligned}$$

The first is the one you'll probably use the least, it involves figuring out an analytic expression of the expected value of second derivative and then plugging in the MLE. The second is a direct substitute for the first: just used the actual second derivatives (not their expectation) at the MLE. The third is sometimes the easiest as you only need the first derivatives. Under our assumptions, these three different estimators are identical asymptotically. In-sample they may be different, but should be very similar. It shouldn't matter which one you pick.

1.4.1 Hypothesis testing

We can now start thinking about hypothesis testing. We'll start with the idea of comparing two nested models $f(y|\theta)$ and $f(y|\theta')$ using the data y . By nested we mean that we can write θ' as some function of the full parameter vector θ , e.g., $h(\theta) = \theta'$. Our intuition here will be to consider the unrestricted log-likelihood that gives us the MLE $\hat{\theta}$ and ask ourselves: Is the log-likelihood that we would get if the null hypothesis is true different enough from the full model that we prefer the full model?

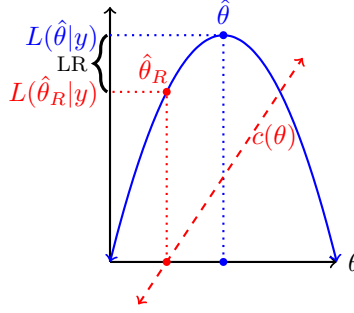
Let $c(\theta) := h(\theta) - \theta'$ then the hypotheses are

$$\begin{aligned}
H_0 : c(\theta) &= 0 \\
H_A : c(\theta) &\neq 0.
\end{aligned}$$

For illustration suppose the log-likelihood function looks like this

The question we're asking of the data is: Is $L(\hat{\theta}_R|y)$ notably worse than $L(\hat{\theta}|y)$? We can imagine something like trying to build a confidence interval around $L(\hat{\theta}|y)$ and asking is $L(\hat{\theta}_R|y)$ in that interval? For example, such an interval could take the form of containing all θ such that

$$CI = \{\theta : L(\hat{\theta}|y) - L(\theta|y) \leq \delta\},$$



where we chose δ to reflect a confidence level.

With that intuition in mind, can start by considering a Taylor series expansion of the likelihood around the unrestricted estimates $\hat{\theta}$. We assumed that L is three times differentiable so we can get

$$L(\theta|y) = L(\hat{\theta}|y) + s(\hat{\theta}|y)'(\theta - \hat{\theta}) + \frac{1}{2}(\theta - \hat{\theta})'[H(\hat{\theta}|y)](\theta - \hat{\theta}) + R(\theta; \bar{\theta}).$$

Here $R(\theta; \bar{\theta})$ is the remainder term with $\bar{\theta}$ in between $\theta - \hat{\theta}$. We can rearrange terms

$$\begin{aligned} 2(L(\hat{\theta}|y) - L(\theta|y)) &= (\theta - \hat{\theta})' [-H(\hat{\theta}|y)] (\theta - \hat{\theta}) - 2R(\theta; \bar{\theta}), \\ &= \sqrt{N}(\theta - \hat{\theta})' \left[-\frac{1}{N} H(\hat{\theta}|y) \right] \sqrt{N}(\theta - \hat{\theta}) - 2R(\theta; \bar{\theta}), \end{aligned}$$

and consider some asymptotics

$$\begin{aligned} R(\theta; \bar{\theta}) &\xrightarrow{p} 0 \\ -\frac{1}{N} H(\hat{\theta}|y) &\xrightarrow{p} I(\theta) \\ \sqrt{N}(\theta - \hat{\theta}) &\xrightarrow{d} N(0, I(\theta)^{-1}) \\ (\sqrt{N}(\theta - \hat{\theta}))' \left[-\frac{1}{N} H(\hat{\theta}|y) \right]^{1/2} &\xrightarrow{d} N(0, 1) \\ (\sqrt{N}(\theta - \hat{\theta}))' \left[-\frac{1}{N} H(\hat{\theta}|y) \right] (\sqrt{N}(\theta - \hat{\theta})) &\xrightarrow{d} \chi_J^2 \\ 2(L(\hat{\theta}|y) - L(\theta|y)) &\xrightarrow{d} \chi_J^2, \end{aligned}$$

where J is the number of elements in θ that are not equal to $\hat{\theta}$. In other words, J is the number of restrictions implied by the hypothesis. In most cases we can count up the number of equal signs in the null hypothesis and that will be J .

Property A7 Under Assumptions A1-A8, the null hypothesis $H_0 : c(\theta) = 0$ implies an LR

statistic

$$2(L(\hat{\theta}|y) - L(\hat{\theta}_R|y)) \xrightarrow{d} \chi_J^2.$$

where $\hat{\theta}_R$ is found by maximizing the restricted likelihood.

$$\begin{aligned} \hat{\theta}_R &= \operatorname{argmax}_{\theta} L(\theta|y) \\ \text{s.t. } c(\theta) &= 0. \end{aligned}$$

The LR test has a nice intuition, but it does require comparing two different fitted models. Fitting a second model may be trivial or it could be costly. So what can we do with just a single model? Well we know that MLEs are asymptotically normal, which means that functions of MLEs are also asymptotically normal (because they are themselves MLEs). As a result, all familiar asymptotically justified hypothesis tests you used in 602 are on the table.

Property A8 *Under Assumptions A1-A8, the null hypothesis $H_0 : A\theta = b$ implies a Wald test*

$$(A\hat{\theta} - b)' [A\widehat{\text{avar}}(\hat{\theta})A']^{-1} (A\hat{\theta} - b) \xrightarrow{d} \chi_J^2.$$

For example, so suppose we want to test $J \geq 1$ linear hypotheses such that

$$\begin{aligned} H_0 &: A\theta = b \\ H_A &: A\theta \neq b \end{aligned}$$

What would our test statistic be? Like before we start by assuming a true null hypothesis such that

$$\begin{aligned} A\hat{\theta} - b &= A\hat{\theta} - A\theta \\ &= A(\hat{\theta} - \theta) \\ \sqrt{N}(A\hat{\theta} - b) &= A\sqrt{N}(\hat{\theta} - \theta) \\ \sqrt{N}(A\hat{\theta} - b) &\xrightarrow{d} N(0, AI(\theta)^{-1}A') \\ [A\Omega A']^{-1/2}\sqrt{N}(A\hat{\theta} - b) &\xrightarrow{d} N(0, I_J) \\ N(A\hat{\theta} - b)'[AI(\theta)^{-1}A']^{-1}(A\hat{\theta} - b) &\xrightarrow{d} \chi_J^2 \\ N(A\hat{\theta} - b)'[AI(\hat{\theta})^{-1}A']^{-1}(A\hat{\theta} - b) &\xrightarrow{d} \chi_J^2 \\ (A\hat{\theta} - b)' \left[A \frac{1}{N} I(\hat{\theta})^{-1} A' \right]^{-1} (A\hat{\theta} - b) &\xrightarrow{d} \chi_J^2 \\ (A\hat{\theta} - b)' [A\widehat{\text{avar}}(\hat{\theta})A']^{-1} (A\hat{\theta} - b) &\xrightarrow{d} \chi_J^2 \end{aligned}$$

Note that if $J = 1$, we have the simpler z test where

$$\frac{\hat{\theta}_j - b}{\sqrt{\widehat{\text{avar}}(\hat{\theta}_j)}} \xrightarrow{d} N(0, 1).$$

Not all interesting tests are linear however. Sometimes we may have a length J nonlinear hypothesis or transformation in the form of a continuously differentiable function $c(\theta)$.

In order to make progress here, we will need to know the distribution of $c(\hat{\theta})$. This is a question about a distribution of a nonlinear function of a normally distributed random variable.

We can again start with a mean-value representation

$$c(\hat{\theta}) = c(\theta^*) + D_{\theta}c(\bar{\theta})(\hat{\theta} - \theta^*)$$

Applying all the same arguments as before we get:

Property A9 *Under Assumptions A1-A8, the distribution of $c(\hat{\theta})$, where c is continuously differentiable, is asymptotically normal with:*

$$\sqrt{N}(c(\hat{\theta}) - c(\theta)) \xrightarrow{d} N\left(0, D_{\theta}c(\theta)I(\theta)^{-1}D_{\theta}c(\theta)'\right).$$

Notice that this is the **delta method** expression for the variance of $c(\hat{\theta})$. This result gives us a firm basis for using the delta method to produce standard errors for transformations of MLE estimates. Note that this also complements an earlier result that $c(\hat{\theta})$ is itself the MLE of $c(\theta)$.

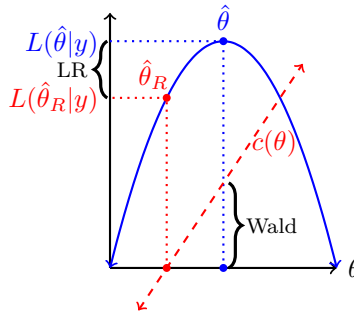
So if $c(\hat{\theta})$ is a singleton then we can construct a z test based on this. When $c(\hat{\theta})$ is vector-valued, we can apply the logic from the linear hypothesis test we can get to a usable test statistic for the hypothesis that $c(\theta) = 0$ and

Property A10 *Under Assumptions A1-A8, the null hypothesis $H_0 : c(\theta) = 0$ implies a Wald statistic*

$$c(\hat{\theta})' \left[D_{\theta}c(\hat{\theta})\widehat{\text{avar}}(\hat{\theta})D_{\theta}c(\hat{\theta})' \right]^{-1} c(\hat{\theta}) \xrightarrow{d} \chi_J^2.$$

Going back to our picture, we can see how the Wald test differs from the LR test.

Let's consider an example with 10 iid observations from a $N(1, 1)$. In this case, suppose we want to test the hypothesis that $\mu = 0.5$. We can do this in several ways now Recall that the MLE for μ is the sample mean and the MLE for σ^2 is $N^{-1} \sum (y_i - \hat{\mu})^2$. For the LR test we



will compare the unrestricted likelihood evaluated at the MLEs

$$L(\theta|y) = \sum_{i=1}^N \log f(y_i|\theta),$$

to the likelihood evaluated at the θ_R . Here θ_R is?

$$\hat{\mu}_R = 0.5$$

$$\hat{\sigma}_R^2 = \frac{1}{N} \sum_{i=1}^N (y_i - 0.5)^2.$$

```
set.seed(1)
y <- rnorm(10, 1,1)
mu.mle <- mean(y)
s2.mle <- mean((y-mu.mle)^2)
s2.mle.R <- mean((y-0.5)^2)

## LR test
L.unrestricted <- sum(dnorm(y, mu.mle, sqrt(s2.mle), log=TRUE)) #at hat(theta)
L.unrestricted

## [1] -11.18548

L.restricted <- sum(dnorm(y, 0.5, sqrt(s2.mle.R), log=TRUE)) #at hat(theta)_R
L.restricted

## [1] -13.92272

LR.test <- 2*(L.unrestricted-L.restricted)
LR.test

## [1] 5.474474
```

```
pchisq(LR.test, lower=FALSE, df=1)
```

```
## [1] 0.01929617
```

```
# Wald
```

```
Dtheta <- cbind(-mu.mle/s2.mle+y/s2.mle,  
               -1/(2*s2.mle)+(y-mu.mle)^2 / (2*s2.mle^2))  
OPG <- t(Dtheta) %*% Dtheta  
vcov <- solve(OPG)
```

```
wald.stat <- (mu.mle -0.5)/sqrt(vcov[1,1])  
wald.stat
```

```
## [1] 2.605179
```

```
pnorm(abs(wald.stat), lower=FALSE)*2
```

```
## [1] 0.009182631
```

```
## or
```

```
theta.hat <- c(mu.mle, s2.mle)  
A <- c(1,0)  
b <- 0.5  
wald2 <- (A%*% theta.hat -b) %*% solve(A %*% vcov %*% A) %*% (A%*% theta.hat -b)  
wald2
```

```
##           [,1]
```

```
## [1,] 6.786957
```

```
pchisq(wald2, lower=FALSE, df=1)
```

```
##           [,1]
```

```
## [1,] 0.009182631
```

1.5 Misspecification of the DGP

A quick time out may also be in order here. Notably, we have relied on the assumption that we know the density f that generates the data. This is, of course, absurd. We never really know the true DGP, but it puts us in a case where we return to the age-old truism that “all models are wrong, but some are useful.” Given that we’re admitting our ignorance regarding the model, we have two choices:

1. If all we're interested in is average marginal effects and all the quantities of interest are observable, then OLS may be more attractive, given its somewhat weaker assumptions. However, the kinds of questions we can consider may be severely limited if we only allow ourselves to search for one-off, black-box average treatment effects in reduced form.
2. We accept that more often than not we're actually choosing a model that may be more or less **close** to the true DGP. Let's see where we can go with this.

Ok, let's suppose that the true model is $g(y|\theta)$. What happens if we fit the data using a model of $f(y|\beta)$ instead? To make progress here, we'll need to consider a measure of the "gap" between models. We don't really know how β and θ relate to each other, so instead of looking for the distance between them, we'll need to look at the divergence between f and g ,

$$KL(g; f) = \int g(y|\theta) \log \frac{g(y|\theta)}{f(y|\beta)} dy.$$

This is known as the **Kullback-Liebler (KL) divergence** from f to g . If g is the true model then we get

$$KL(g; f) = E \left[\log \frac{g(y|\theta)}{f(y|\beta)} \right].$$

A few things to point out

1. $KL(g; f) \geq 0$. To see this note that

$$KL(g; f) = E \left[-\log \frac{f(y|\beta)}{g(y|\theta)} \right] \geq E \left[1 - \frac{f(y|\beta)}{g(y|\theta)} \right] = \int g(y|\theta) dy - \int f(y|\beta) dy = 1 - 1 = 0.$$

2. If $f = g$ then $KL(g; f) = 0$, which is to say there is no divergence. This is the only case where the divergence is 0.

Before going on, let's consider two examples. First suppose that f and g are both normal

with means μ_f and μ_g , but the same variance. Then we have

$$\begin{aligned}
\log \frac{g(\theta)}{f(\beta)} &= \log \frac{\exp\left(-\frac{(y-\mu_g)^2}{2\sigma^2}\right)}{\exp\left(-\frac{(y-\mu_f)^2}{2\sigma^2}\right)} \\
&= \frac{\mu_f^2 - \mu_g^2 - 2y(\mu_f - \mu_g)}{2\sigma^2} \\
KL(g; f) &= \mathbb{E} \left[\log \frac{g(\theta)}{f(\beta)} \right] \\
&= \frac{\mu_f^2 - \mu_g^2 - 2\mathbb{E}[y](\mu_f - \mu_g)}{2\sigma^2} \\
&= \frac{(\mu_f - \mu_g)^2}{2\sigma^2}.
\end{aligned}$$

In this case the divergence from f to g is proportional to the squared difference in means. Also note that in this case $KL(g; f) = KL(f; g)$, but that will rarely be the case.

Second, suppose that g is log-normal, but everything else is the same, then

$$KL(g; f) = \frac{\mu_f^2 - 2\mu_f \exp(\mu_g + \sigma^2) - \sigma^2 + \exp(2\mu_g + 2\sigma^2)}{2\sigma^2}$$

which will generally not be the same as $KL(f; g)$

OK, so let's consider then what happens when we fit the wrong model. First of all, we end up with our estimate is

$$\hat{\beta} = \operatorname{argmax}_{\beta} \frac{1}{N} \sum_{i=1}^N \log f(y_i|\beta).$$

From our discussion of consistency we know that this likelihood function converges to

$$\frac{1}{N} \sum_{i=1}^N \log f(y_i|\beta) \xrightarrow{P} \mathbb{E}_g[\log f(y|\beta)].$$

Note that g though, because this expectation is conditional on the true DGP g . What can we say about this quantity?

$$\begin{aligned}
\mathbb{E}_g[\log f(y|\beta)] &= \mathbb{E}_g[\log f(y|\beta)] - \mathbb{E}_g[\log g(y|\theta)] + \mathbb{E}_g[\log g(y|\theta)] \\
&= \mathbb{E}_g[\log g(y|\theta)] - \mathbb{E}_g \left[\log \frac{g(y|\theta)}{f(y|\beta)} \right] \\
&= \mathbb{E}_g[\log g(y|\theta)] - KL(g; f).
\end{aligned}$$

The first term doesn't depend on β , so the MLE $\hat{\beta}$ associated with maximizing $\mathbb{E}_g[\log f(y|\beta)]$

is the one that minimizes $KL(g; f)$. We'll take this as our next property

Property A11 *Let $y_1, \dots, y_N$ be an iid sample from distribution $g(y|\theta)$ and let $KL(g(y|\theta); f(y|\beta))$ to have unique minimum $\beta = \beta^*$, then with regularity conditions on g and f , the MLE $\hat{\beta} \xrightarrow{p} \beta^*$*

In words, this means that the MLE obtained by fitting the data to the incorrect model f , will converge to the parameter values that make the incorrect model *as close as possible* to the unknown correct model.

In this sense, we can rest somewhat easy as long as we think the model is close-enough or a reasonable approximation to the real model.

The next question becomes, however, what happens to the variance of the MLE in these cases? We derived the previous variance expression using the assumption of a correct model. So what changes now that we're even more uncertain?

The main thing, we need to think about here is what happens to the expected value when we go to apply the LLN.

$$\begin{aligned}\sqrt{N}(\hat{\beta}_N - \beta^*) &= \left[-\frac{1}{N} H(\bar{\beta}|y) \right]^{-1} \sqrt{N} \left(\frac{1}{N} s(\beta^*|y) \right) \\ \frac{1}{N} H(\bar{\beta}|y) &\xrightarrow{p} E_g [H_i(\beta^*|y_i)] \\ \frac{1}{N} s(\beta^*|y) &\xrightarrow{p} 0 \\ \sqrt{N} \left(\frac{1}{N} s(\beta^*|y) \right) &\xrightarrow{d} N(0, \text{Var}_g(s_i(\beta^*|y_i))) \\ \sqrt{N}(\hat{\beta}_N - \beta^*) &\xrightarrow{d} N(0, E_g [H_i(\beta^*|y_i)]^{-1} \text{Var}_g(s_i(\beta^*|y_i)) E_g [H_i(\beta^*|y_i)]^{-1})\end{aligned}$$

When we were confident in our model choice, we could employ the information equality, which told us that

$$\text{Var}_g(s_i(\beta^*|y_i)) = E_g [H_i(\beta^*|y_i)], \quad \text{if } f = g.$$

We don't have that reassurance anymore so we can't claim this equality. Instead, we will use the sample counterparts of these matrices to construct a sandwich estimator.